ECON 8803: PS2

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This problem is based on the paper by Almond, Chay, and Lee (2005) and is a modified version of a problem set by Kyrill Borusyak. The goal of this problem set is to examine the causal effect of maternal smoking during pregnancy on infant birth weight. The data for the problem set comes from the 1993 National Natality Detail Files for Pennsylvania. Each observation represents a mother-infant match. The data in Stat and CSV formats can be downloaded from Canvas. The TXT file gives variable names and labels. There should be 43 variables in the data and after you are finished with the cleaning steps described below, 114,610 observations. The PDF codebook for the data is also posted and will help you understand the relevant variables.

The data here are "real" and quite imperfect, which will help to simulate the pain of real world data work. While in the real world where you may confront this largely alone, I will provide some hints for working your way through this data and research question.

1 Data cleaning

(a) Fix missing values: in the data set, several variables take on a special value (e.g., 9999) if missing. Tabulate the variables and check the codebook for missing value codes; then replace special values with missing. This has already been done for roughly 2/3 of the variables; the remaining variables you need to check are the last 15, from `cardiac` to `wgain`.

(b) Produce an analysis dataset that drops any observation with missing values and verify it has 114,610 observations. Drop `stresfip`, `birmon`, `weekday`.

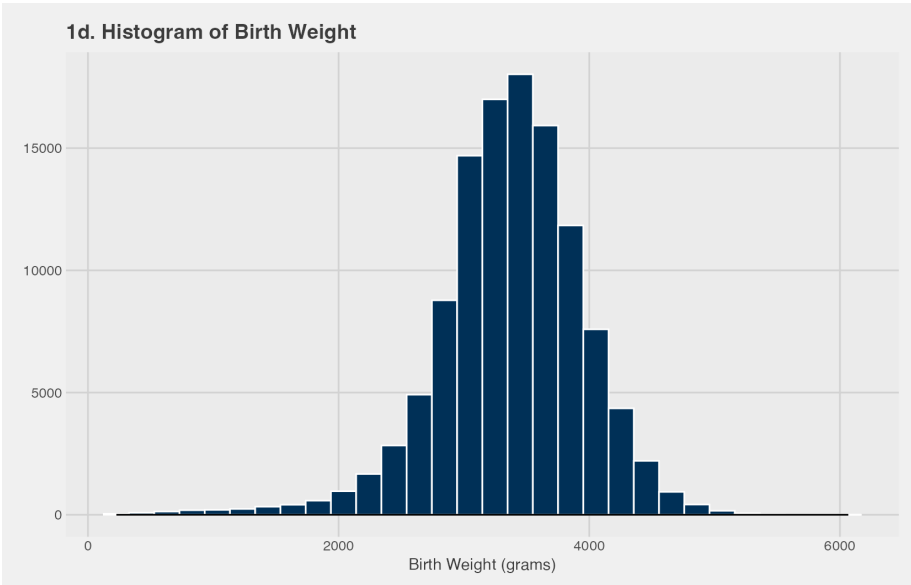
(c) Produce a summary statistics table describing some of the key variables in the final analysis data set. A good summary table generally includes mean, standard deviation, minimum and maximum. Pay attention to how some variables are measured (continuously, categorically, etc.). This may be a large table, feel free to break it up as you feel appropriate.

Summary Statistics: Continuous Variables

	Mean	Median	SD	Min	Max
Mother's Age	27.76	28.00	5.70	12	49
Mother's Education	13.21	12.00	2.27	0	17
Number of Live Births (Now Living)	0.97	1.00	1.15	0	12
Number of Live Births (Now Dead)	1.99	2.00	1.17	1	14
Total Birth Order	2.41	2.00	1.46	1	8
Month Prenatal Care Began	2.50	2.00	1.33	0	9
Total Number of Prenatal Visits	11.15	12.00	3.52	0	49
Interval Since Last Live Birth	3.32	4.00	3.19	0	9
Father's Age	30.06	30.00	6.41	13	78
Father's Education	13.28	12.00	2.33	0	17
Gestation - Detail in Weeks	39.15	39.00	2.44	17	47
Birthweight (g)	3,373.29	3,402.00	585.17	227	6067
Clinical Estimate of Gestation	39.11	40.00	2.06	17	44
Avg Number of Cigarettes Per Day	0.35	0.00	0.86	0	5
Avg Number of Drinks	0.02	0.00	0.23	0	4
Weight Gain	30.36	30.00	11.88	0	98

Summary Statistics: Categorical Variables		
Variable	**N**	**Proportion**
Female Children	114610	0.485
Married Mothers	114610	0.251
Hispanic Mothers	114610	0.036
Hispanic Fathers	114610	0.038
Multiple Births	114610	0.027
Anemic Mothers	114610	0.010
Mothers with Cardiac Disease	114610	0.007
Mothers with Lung Disease	114610	0.007
Diabetic Mothers	114610	0.027
Mothers with Genital Herpes	114610	0.006
Mothers with Chronic Hypertension	114610	0.008
Mothers with Pregnancy-Related Hypertension	114610	0.031
Previous Infants Weighing > 4000g	114610	0.014
Previous Preterm Infants	114610	0.014
Tobacco Use During Pregnancy	114610	0.159
Alcohol Use During Pregnancy	114610	0.010

(d) Generate (and report) a histogram of the birth weight variable. This is our outcome of interest.



2 Preliminary analysis

(a) Compute the mean difference in birth weight by maternal smoking status.

Birthweight by Tobacco Use				
Group	Category	n	Mean	SD
0	Non-Smoker	96344	3,411.62	579.73
1	Smoker	18266	3,171.14	572.08

- The mean difference in birthweight by maternal smoking status is 240.48g.

(b) Is this difference likely to be causal? Why or why not? (Hint: a common piece of compelling evidence is a table checking the balance of several other relevant variables across treatment status.)

- No, we do not think this difference is likely to be causal given the data that we have- the balance table shows that there are systemic

differences between the two groups in terms of the composition. Additionally, the groups are not randomly assigned.

Balance Table by Smoking Status		
Characteristic	Non-Smokers	Smokers
n	96,344.00	18,266.00
mean_brwt	3,411.62	3,171.14
mean_age	28.06	26.17
mean_education	13.44	11.99
pct_married	79.27	51.78
pct_black	2.29	0.38
mean_prenatal_visits	11.25	10.63
pct_hypertension	3.27	2.02
pct_diabetes	2.69	2.68

(c) You decided to use the "selection on observables" identification strategy which involves assuming that maternal smoking during pregnancy is randomly assigned conditional on some observable determinants of infant birth weight. Now it's time to choose those covariates using informed judgment. Classify the covariates into two main categories.

- Potential confounders or "pre-determined" determinants of the outcome: variables that are **not caused** by smoking during the current pregnancy but can either affect both smoking and birth weight **or** proxy for unobserved variables that do
  - We include the following as potential confounders:
  - Maternal Demographics: dmage, dmeduc, mrace3, ormoth, dmar
  - Paternal Demographics: dfage, dfeduc, orfath
  - Pregnancy History: nlbnl, dlivord,totord9, isllb10, csex
  - Prenatal Care: monpre, nprevist, adequacy
  - Substance Use: alcohol, drink5
- Bad controls: variables that can be causally affected by smoking, regardless of whether they are related to birthweight
  - I would exclude the following covariates as bad controls impacted by smoking: dgestat, phyper, preterm, pre4000. Many of these diseases are heavily correlated with or can be caused by smoking.

3 Regression adjustment

Collect all the estimates and standard errors from the regressions and show these in one well-formatted table.

Table 1: Regression results

Specification	**Estimate**	**Standard Error**
Uninteracted Linear Regression	-220.276	4.821
Bad Controls	-205.1	4.063
cigar6	-119.104	11.813
tobacco * diabetes	-222.815	4.849
Oaxaca-Blinder	-240.478	

(a) Use a basic, uninteracted linear regression model to estimate the impact of smoking conditional on the set of pre-determined variables and report your estimates. Interpret the coefficient estimate. Under what circumstances does it identify the average treatment effect (ATE)?

- The coefficient is -221 grams: the average baby born to smoking mothers weighs 221 grams less than babies born to non-smoking mothers with $p < 2 \times 10^{16}$.
- This will identify the average treatment effect when treatment is effectively randomly assigned when controlling for covariates. The ATE holds when the conditional independence holds.

Table 2: 2a. Regression Table

	Dependent variable:
	dbrwt
tobacco	-221.027^{***}
	(4.791)
dmage	-4.529^{***}
	(0.518)
dmeduc	8.656^{***}
	(1.033)
mrace3	-109.731^{***}
	(2.799)
ormoth	-33.090^{***}
	(4.172)

dmar	-66.319^{***}
	(4.861)
dfage	-0.108
	(0.405)
dfeduc	2.933^{***}
	(0.964)
orfath	-17.395^{***}
	(4.112)
nlbnl	128.164^{***}
	(10.478)
dlivord	-82.704^{***}
	(10.440)
totord9	-16.621^{***}
	(2.110)
isllb10	18.344^{***}
	(0.668)
monpre	18.067^{***}
	(1.693)
nprevist	30.236^{***}
	(0.620)
adequacy	62.395^{***}
	(4.925)
alcohol	108.979^{***}
	(34.766)
drink5	-88.715^{***}
	(14.801)
csex	-124.636^{***}
	(3.286)
Constant	$3{,}351.929^{***}$
	(23.029)

Observations	114,610
R^2	0.097
Adjusted R^2	0.097
Residual Std. Error	555.963 (df = 114590)
F Statistic	651.532^{***} (df = 19; 114590)
Note:	*p<0.1; **p<0.05; ***p<0.01

(b) For this part only, add to the specification in question 3(a) some "bad controls." Check if your estimate changes and discuss the direction of the change.

- After adding back in `dgestat`, `phyper`, `preterm`, and `pre4000`, the new coefficient is -207.1g. I think this is because tobacco use directly affects all three of those variables, thereby reducing birthweight. The direction of the change makes sense.

(c) Now, add to the specification in question 3(a) the variable `cigar6`. What happens to the coefficient on smoking status? Why do you think that happens?

- The coefficient reduced all the way to -119.1g. This is because `cigar6`, which is the recode variable for the average number of cigarettes per day, directly overlaps with tobacco usage.

(d) Next, explore treatment effect heterogeneity by interacting smoking status with the indicator for whether the mother has diabetes. Use the same set of controls that you used in 3(a). Provide an interpretation of the interaction term (2-3 lines max).

- When interacting tobacco usage with diabetes status, the estimate for the interaction term is 64.249. Here, the coefficient for tobacco at -222.815 and for diabetes alone at 40.022. The combined effect for diabetic mothers who smoke is -156.844 (-222.815 + 65.249).

(e) Now, use the Oaxaca-Blinder logic to calculate the ATE (i.e., add interactions of covariates with smoking status). Briefly describe the exact steps you follow. Does your result differ substantially from the one in 3(a)? Intuitively,what does this tell us about treatment effect heterogeneity?

- We first ran a linear regression interacting smoking with all covariates. After creating new tables, one for smokers and one for non-smokers, we used R's `predict` function to find the means of each dataset.
- The average treatment effect between smoking and non-smoking mothers was -240.478, suggesting heterogeneity that varies by covariates.

4 Propensity score methods

Collect all the ATE estimates and show these in one well-formatted table.

Table 3: ATE Across Methods

Specification	**Estimate**
Linear Probability Model	-220.365
Logit	-220.379
Blocking On Propensity Score	-226.367
Hajek IPW	-223.348

(a) Use a LPM to estimate propensity score using the covariates identified in 2c. Now, use the estimated p-score in an OLS regression of birthweight on smoking status. What do you observe? Why is that?

- The coefficient estimate (-221g) is equivalent to the basic linear regression model because propensity score captures the variance from each covariate into a tidy package.

(b) Estimate the p-score using a logit specification without nonlinear terms and interactions. Discuss which covariates appear most predictive of maternal smoking and whether this matches your expectation (please do not use any canned propensity score commands here).

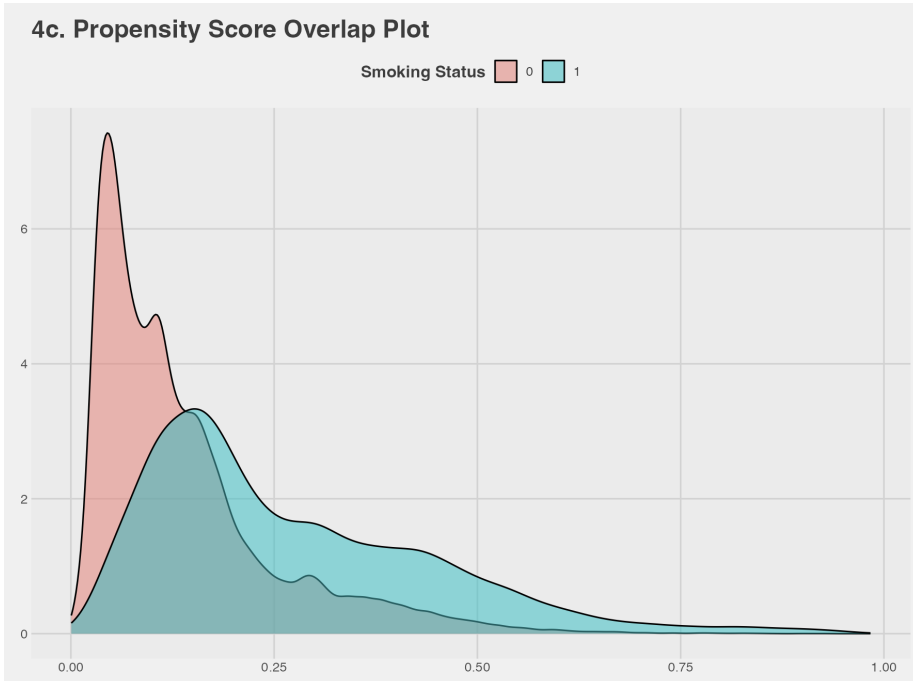
- Two covariates with the highest coefficients were mother's marital status (1.233) and alcohol (1.281). On the other side, mother's race (-0.538) and Hispanic origin (-0.418) also had substantial values. Alcohol in particular makes sense as having an outsized impact on smoking status as they are heavily correlated.

(c) For the p-score estimated using the logit specification, how good is the overlap between the treated and untreated groups?

- There seems to be substantial overlap between non-smokers and smokers. However, a decent amount of non-smokers at low p-scores do seem to escape overlap.

(d) Now estimate the average treatment effect of maternal smoking on birthweight using OLS regression and controlling for the propensity score. Interpret the treatment effect. State clearly the assumptions under which your estimate has a causal interpretation.

- Under the conditional independence assumption where the outcomes are independent on treatment conditioned with the propensity score, smoking mothers see a 220.379g reduction in birthweight compared to nonsmoking mothers.



(e) Estimate an ATE by the blocking-on-propensity -score method. To do this, divide the data into 10 groups defined by equally sized ranges of the estimated propensity score. Use the blocking estimator from lecture to estimate the ATE of interest.

- Using the blocking-on-propensity-score method, we estimate the ATE of interest as -226.367g.

(f) Estimate the ATE using the Hajek propensity score reweighting (IPW) estimator. You can compute this manually or run a weighted-regression as discussed in the lecture:

$$\widehat{ATE} = \left(\frac{\sum_i \frac{D_i Y_i}{\hat{p}(X_i)}}{\sum_i \frac{D_i}{\hat{p}(X_i)}} \right) - \left(\frac{\sum_i \frac{(1-D_i)Y_i}{1-\hat{p}(X_i)}}{\sum_i \frac{(1-D_i)}{1-\hat{p}(X_i)}} \right)$$

- We computed this manually and found:

$$\widehat{ATE} = ATE_{treated} - ATE_{control} = -223.348g$$

This aligns nicely with our original estimate of -221g.

5 Doubly robust methods

(a) Implement p-score matching with regression adjustment (you can use canned commands or implement this procedure manually). Briefly describe the exact steps you used.

- We used the `MatchIt` package in R to use nearest neighbor to match on propensity score. After producing the matched dataset, we ran our linear regression of birthweight and found the coefficient to be -223.378g, consistent with our previous findings.

(b) Implement a double-selection LASSO technique (canned or manual). Use a flexible specification with some polynomials and interaction terms (defining non-linear terms for the use in LASSO may require some thinking).

i. Describe briefly the estimation steps, even if you are using a canned command. Briefly state and justify the specific options you chose in the canned command.

- Using rlassoEffect, we used the covariates from the original regression and added both polynomial and interaction terms for key variables. The first step of LASSO runs a regression of the treatment, smoking status, on all 27 covariates. The second regresses the outcome, birthweight, on the covariates.

Table 4: Double-Selection LASSO

	Estimate	Std. Error	t value	$\Pr(>\|t\|)$
Tobacco	-220.365	4.782	-46.09	$< 2 \times 10^{-16}$ ***

ii. How many covariates did you start with? How many covariates are selected in each LASSO step and how many overlap?

- We started with 27 covariates. The treatment model selected 26, and the outcome model selected 27. The only excluded variable was polynomial maternal marital status (coded as $dmar^2$).

6 Note

Write a short note comparing your estimates from regression methods, propensity score methods, and double-selection. Comment on whether you think the treatment effect estimates are stable across different methods. Comment on the overarching assumption made across all these methods (restrict discussion to half a page).

Across all of the techniques we used, our estimate of the effect of smoking was stable across methods:

Technique	**Estimate**	**SE**
Uninteracted Linear Regression	-221.027	4.791
Linear Propensity Score	-221.027	4.983
ATE Using OLS Controlling on P-Score	-220.379	4.997
P-Score Matching + Regression Adjustment	-223.378	5.857
Double Selection LASSO	-220.365	4.782

The key assumption across all of the methods is the conditional independence assumption (aka selection on observables): $Y_i(1).Y_i(0) \perp D_i|X_i$. This assumes no unobserved confounders. However, realistically, there are a variety of social and environmental factors that are not captured in this data that could contribute to heterogeneity. Factors such as income, geography, and genetic predisposition both for smoking and for maternal and fetal outcomes are some potential omitted variables that would help this causal estimate be more robust.